Friday 13 November | 10:15–11:00

How protomodular is your favourite category?

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The aim of this talk is to investigate a relative notion of protomodularity. We say that a finitely complete category C satisfies the $\mathcal{M}$-*version of protomodularity* when the change-of-base functors (of the fibration of points) are conservative with respect to a class $\mathcal{M}$ of morphisms of C. When $\mathcal{M}$ is the class of all (mono)morphisms of C, this notion is precisely that of an "ordinary" protomodular category. We analyse which conditions on $\mathcal{M}$ are needed to obtain the relative $\mathcal{M}$-versions of some well-known properties of protomodular categories, such as the Split Short Five Lemma (when C is also pointed) and the relation with Mal'tsev categories. We then investigate the $\mathcal{M}$-version of protomodularity for different subclasses of (mono)morphisms in C such as: isomorphisms, strong monomorphisms, split monomorphisms, protosplit monomorphisms (i.e. kernels of split epimorphisms) and fully faithful monomorphisms in an Ord-enriched context. This leads to diverse examples of categories satisfying the $\mathcal{M}$-version of protomodularity for different choices of $\mathcal{M}$. None of these examples are protomodular categories in the usual sense, yet each retains a protomodular "flavour", depending on the choice of $\mathcal{M}$.

The authors acknowledge partial financial support by CMUC (https://doi.org/10.54499/UID/00324/2025), funded by the Portuguese Foundation for Science and Technology (FCT), grants UID/00324/2025 and UID/PRR/00324/2025. The second author also acknowledges financial support by CIDMA (https://ror.org/05pm2mw36), funded by the Portuguese Foundation for Science and Technology (FCT, https://ror.org/00snfqn58), grants UID/04106/2025 (https://doi.org/10.54499/UID/04106/2025) and UID/PRR/04106/2025.

Keywords: protomodular categories; Mal'tsev categories; categories enriched in the category of preorders
Subject area: Categorical Algebra

References

[1] M. M. Clementino, D. Rodelo, *How protomodular is your favourite category?*, arXiv:2610.09739, DMUC preprint.